PA194NS and PA194SUMS
Two new 194-state consistency machines

Construction and verification report

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Abstract

No machine with fewer than 194 working states was established. The successful fallback consists of two explicit, two-symbol, one-tape, blank-input machines with 194 working states: PA194SUMS and PA194NS. They preserve the original sum-based Shift guard while reversing the operands of its destructive comparison. This eliminates a register-transfer loop. PA194NS additionally uses the no-push successful modus-ponens protocol. A new one-bit history fold absorbs the extra control state needed by the replacement trailer.

The construction includes exact tables, dense state maps, a finite local-fold certificate, universal symbolic source checks, an exact blank-startup check, and a PA-verifiable finite-trace equivalence to the appropriate calibrated parent source. The logical calibration is inherited from the repository's separate-diagonal-head PA calculus. No newly compiled Lean or Rocq endpoint is claimed. Neither transition graph occurs in the audited current corpus, which has 51 different byte presentations in 28 graph classes, including rejected diagnostic candidates.

The finite certificates were replayed in independent JavaScript/V8 implementations and then in the supplied Python and Node.js programs. The source bundle includes the complete tables, proof references, graph audit, verification receipts, and reproducible build scripts.

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1 Result, machine convention, and scope

A state count in this report excludes the distinguished halt state. Each machine has a two-way infinite tape, alphabet $\{0, 1\}$, blank symbol 0, one head, and only left or right moves. Dense working states are $0, \ldots, 193$; the initial state is 0 and the halt label is 194. There are exactly 388 transition rows. In inherited native labels the initial state is 74 and halt is 254.

Write H_M for the arithmetical assertion that the literal table M has a finite halting computation from its all-zero tape. The intended theorem is

$$\mathsf{PA} \vdash \neg H_M \longleftrightarrow \mathrm{Con}(\mathsf{PA}).$$

This is an internal proof-translation statement, not an inference from an external belief that the machine runs forever. The report supplies the new finite-trace transport and identifies the inherited source-calibration and arithmetic-engine dependencies explicitly.

Machine	States	Source sites	Outcomes	Startup steps
Primary PA194	194	3673	4780	58154
PA194SUMS	194	3670	4776	58004
PA194NS	194	3651	4751	58004

The shorter source and startup do not imply a global runtime improvement: the changed comparator has different intermediate counter trajectories.

The novelty is a changed register protocol and a changed literal transition graph. It is not a new arithmetic engine, a new logical basis, or a claim of an entirely unrelated proof enumerator. In particular, the existing cached-head symmetric design PA194CS uses a different guard/protocol. The new SUMS/NS controllers retain the original sum guard and its arbitrary-store meaning.

2 Comparison of the supplied designs

2.1 Inventory and deduplication

The comparison was pinned to the commit on the title page, rather than to a moving `main`. It covered 41 filed research packets under the PA research tree and all five ZIP archives then present in `docs/incoming`. The complete PA-tree census found 2,437 paths ending in `.tm`, representing 620 distinct Git blobs; 619 of those blobs were recognized as complete numeric binary tables. The filed-documents subset has 2,423 paths and 618 blobs, of which 617 are recognized complete tables at 2,420 paths. The other 47 top-level PA directories add 14 paths and two new table byteforms: upstream 196 and 209 states. These counts include old machines, compiler intermediates, and rejected experiments.

The tracked part contributed 15 byte presentations of 194-state tables in eight graph classes. The five incoming archives contributed 36 new byte presentations in 20 further classes after deduplication: a total of 51 presentations in 28 classes. Diagnostic tables were retained in this novelty comparison so that a rejected earlier graph could not be mistaken for a new design. The small six-state and nine-state synthetic tables in the corpus are test/bypass examples, not calibrated PA machines.

Six retained historical ZIP paths identify four distinct outer archives. Recursively following their embedded ancestors gives 14 distinct archive blobs. Their complete numeric tables have 209–222 states; none adds a 194-state-or-smaller graph. The Joint209 subtree has 50 table occurrences and 40 byteforms, the other 209 subtree 115 occurrences and 50 byteforms, and the U212combined subtree 44 occurrences and 35 byteforms. These overlapping subtree counts must not be summed as a deduplicated global total. A supplement contains no transition table. Comment-prefixed headers were stripped for parsing but retained in the exact byte audit.

Graph comparison allowed arbitrary state renaming, global left/right reflection, and global interchange of tape symbols. Candidate roots with a spanning forward graph were considered. The new graphs remain absent even under these more permissive transformations. Byte differences and different dense renumberings were not counted as new architectures.

The packet ledger and graph-class manifest are included with the sources. This is an inventory and a construction audit, not a claim that every historical manuscript has been independently formalized.

2.2 What changes across the families

Family	Principal mechanism	Comparison and consequence
Historical 209–222	Two distinct 209 constructions, a 209 supplement, a 212 entry-bypass construction, and embedded ancestors	The recursive archive audit closes the older-reference scope. These contribute no 194-state graph; intermediate tables extend through 222 states.
Unary-power / U208	Three-cell and contextual folds; variations of entry and history handling	The early 208-state papers contain genuinely different local presentations but no verified smaller endpoint in the reported reduction searches.
PA199 / PA196	Diagonal vacuity; a bundled or separate diagonal clause; AX grammar and register allocation	Source shortening alone is not a state bound. Some smaller source trees compile to larger machines.
PA195	More effective AX allocation, original-guard and cached-head protocols, combined diagonal gate	Distinguish logical protocol changes from alternative physical quotients of the same source.
Primary PA194	67 AX-reader controls and seven history folds	The reference literal table used here. The AD companion reallocates registers; another implementation has a 68/8 budget.
Cached C / CS / CZ	Cache-based guard, operand symmetry, zero jumps	Several deliveries duplicate C exactly. CS already uses symmetry, but with a different Shift interface from the original sum guard.
N	Successful MP omits a zero stack push	Shorter source; proof compilation must explicitly reserve stack entries formerly supplied by the machine.

R / F / A / X	Register arrangements, command variants, and combinations	The shared X/CSZ graph occurs in multiple incoming packages; it is not three independent discoveries.
Radix family	Formula-constructor coefficients 5/6; a coefficient-4 decoder-layout companion	The stack remains unscaled Cantor pairing, and the tuple enumerator and command scheduler are unchanged.
J / D and siblings	Zero-jump changes and diagonal/gate changes	Include accepted primary tables and exploratory grammar variants separately.
Q	Curried-premise exchange in a quantifier-related head	Reordering axiom arguments requires an actual proof translation, not a free syntactic permutation.
SUMS / NS, this work	Original sum guard, reversed destructive comparison; NS also removes the MP zero push	The new fold gives 194 controls in both cases. Neither graph is in the 28-class audit.

For the radix-k formula variants,

$$I_k(a,b) = kp(a,b), \quad M_k(a,b) = kp(a,b) + 1, \quad U_k(i,a) = kp(a,i) + 2, \quad E_k(a,b) = kp(a,b) + 3.$$

The coefficient-5 and coefficient-6 machines change these formula constructors consistently in AX and MP. Their logical translation recodes both formulas and variable labels recursively:

$$F_k(0) = 0, \qquad F_k(4p(a,b) + r) = kp(F_k(a), F_k(b)) + r$$

on nonzero constructor codes. Labels must be recoded to preserve the diagonal head, where a formula code is also a binder label. This map does not preserve numerical order: $40 < 44$ but $F_5(40) = 75 > 55 = F_5(44)$. The restricted-implication dominance condition therefore needs the packet's separate proof-translation argument, rather than an assumption of order preservation. Their stack pairing, tuple enumeration, and scheduler remain unchanged. The coefficient-4 companion has the original resolved source graph and a different decoder arrangement. The new contextual multiplication-decoder sharing gives the same $135 + 67 - 8 = 194$ physical budget.

The most reusable idea from the ZFC material is to optimize source layout, register allocation, and history-sensitive state sharing together. It does not supply a free PA-calibration theorem for a different checker. Likewise, ConciseIPRA supplies ideas about concise encodings and primitive recursive proof checking, but substituting a search for an iPRA contradiction would require a new PA-provable refutation translation in both directions. This report makes no identification of Con(iPRA) with Con(PA).

3 Exact construction

3.1 A nine-row patch specification

Let M be the primary native table distributed in `pa194_conpa_2026-11-13/construction/tables/pa194.native.tm`. Its SHA-256 is

```
e0a7397cf2134e8e7f58a8b1de2f7ff90b74ededd54c912f6ba5b7db48dabe54
```

To construct PA194NS, apply exactly the following changes. An entry (w, D, n) means write w, move D, and enter n.

Control	Read	Old action	New action
28	0	$(0, R, 41)$	$(0, R, 330)$
38	1	$(1, R, 20)$	$(1, R, 30)$
41	0	$(0, R, 24)$	delete
41	1	$(0, R, 25)$	delete
70	0	$(0, R, 41)$	$(0, R, 330)$
164	1	$(1, R, 98)$	$(1, R, 114)$
310	1	$(1, R, 62)$	$(1, R, 330)$
330	0	absent	$(0, L, 31)$
330	1	absent	$(1, L, 46)$

Every other row is byte-for-byte inherited. PA194SUMS is the same patch with row $(164, 1)$ left unchanged. Thus both delete native state 41 and add state 330, preserving the total state count.

Dense labels are obtained by breadth-first discovery from native 74, visiting the read-0 edge before read-1 and assigning halt 254 the dense label 194. The shipped label arrays give the native label of each dense index, with halt last. All 194 working states are graph reachable.

3.2 Source modification

The pinned `PA196/Program.lean` supplies the source DAG; its `x67` subtree is replaced by the primary PA194 AX grammar. The logical-to-physical allocation remains

$$(N, T, W, A, D, H, X, Y, L) \longmapsto (6, 2, 0, 4, 3, 7, 8, 5, 1).$$

For SUMS the source changes are

```
node168 = (node37, node56)
node171 = (node170, (leaf inc 5, leaf continueTo 3))
```

For NS also set

```
node163 = (node6, node54)
```

The latter is `node54`, not `node48`. All other named nodes and the AX grammar stay unchanged.

The first 3,668 resolved instructions of SUMS coincide with primary PA194. The old tail decrements X in a loop, increments H, and finally increments X before returning to the comparator. The replacement has only `inc H; jump 0`. The analogous NS prefix has length 3,649. It is identical to its no-push parent's prefix; only that final transfer tail changes. Prefix identity here refers to resolved instructions, not to identical binary source-address strings.

4 Why reversing the comparison is correct

Lemma 1 (Destructive equality)**.** *The common first eleven source instructions halt exactly when $H = X$. If $H \neq X$, they terminate at instruction 11 with $H = X = 0$ and every other register unchanged.*

Proof. The comparator decrements the two natural counters in lockstep. If both reach zero together it takes the halt branch. If exactly one reaches zero first, the remaining loop clears the other counter and exits. Each loop has a decreasing natural counter, so no nonhalting behavior is introduced inside the block. This is a PA-provable induction on the initial counter sum. □

Lemma 2 (Original sum-guard symmetry)**.** *The old and new Shift blocks have the same halting decision and the same complete nonhalting register store for arbitrary incoming $(T, H, X) = (t, h, x)$.*

Proof. The common prefix first transfers T into X, and then transfers H into both T and X. Put $k = x + t + h$. At this point

$$(T, H, X) = (h, 0, k).$$

The old tail transfers X into H and increments X. The new tail increments H. Their respective comparator inputs are therefore

$$(h, k, 1) \qquad \text{and} \qquad (h, 1, k).$$

By the preceding lemma both halt exactly when $k = 1$. Otherwise both leave $(T, H, X) = (h, 0, 0)$, with every other register identical. All local loops terminate on natural stores. In particular, the statement does not depend on assuming the scratch register X was initially zero. At command boundaries, where $X = 0$, the guard is exactly $t + h = 1$. □

Theorem 1 (Finite-trace transport)**.** *PA proves that SUMS halts if and only if the primary PA194 source halts. It proves that NS halts if and only if the no-push PA194N source halts.*

Proof. Decompose a source computation into unchanged instructions and complete visits to the changed finite block. Replace each old block by its new counterpart, or conversely. The preceding lemma gives identical nonhalting exits and matching halting decisions. The finite blocks terminate with positive duration. Induction on the number of boundary visits yields a primitive-recursive transformation of finite halting traces in each direction. Determinism and positive progress also exclude a spurious halt hidden between simulated boundaries. All these are finite arithmetic statements and inductions available in PA. □

This is an equivalence of halting statements, not a claim that the two machines have identical intermediate configurations or equal running times.

5 The physical quotient and its checks

5.1 The new history fold

Before quotienting, the new parent has 202 controls: 135 fixed controls and the same 67 AX-reader controls. The inherited seven AX folds remain; a new eighth fold identifies old control 41 with the new control 330. The raw actions are

$$41 : (0, R, 24)/(0, R, 25), \qquad 330 : (0, R, 63)/(1, L, 73).$$

Every incoming edge to old 41 has just written 0 and moved right. Every incoming edge to new 330 has just written 1 and moved right. Thus the immediately preceding tape bit records which raw control was entered. The retained 330 uses

$$330 : (0, L, 31)/(1, L, 46).$$

Its complete four-case certificate is

History bit	Current bit	Three-step path	Effective action
0	0	$330, 31, 21, 24$	$(0, R, 24)$
0	1	$330, 46, 61, 25$	$(0, R, 25)$
1	0	$330, 31, 47, 63$	$(0, R, 63)$
1	1	$330, 46, 62, 73$	$(1, L, 73)$

The histories are restored and the effective write, exit head, and exit control agree exactly with the raw transition. Only the history and current cell are inspected. The certificate was checked against the final simultaneous quotient, not against an isolated hypothetical fold.

All eight folds together have 32 history/read cases and 26 incoming rows. Their replay lengths are ten cases of one step, twenty cases of three steps, and two cases of five steps. The resulting count is

$$202 - 8 = 194.$$

The unchanged 23 arithmetic-box controls account for 46 literal rows. No new arithmetic-box lemma is silently assumed.

5.2 Universal source contracts

A source instruction is checked on a symbolic complete register file, with an affine gap rather than a bounded finite sample. Decrement instructions have separate zero and positive outcomes. Increment, jump, and halt have their corresponding complete contracts. Every successful simulation requires positive physical progress.

Machine	Sites	Outcomes	Total symbolic steps	Maximum
Primary positive control	3673	4780	1479348	497
SUMS	3670	4776	1485452	499
NS	3651	4751	1479348	499

All listed outcomes pass. NS has 1,449 increments, 1,100 decrements, 1,101 jumps, and one halt. SUMS has 1,456 increments, 1,106 decrements, 1,107 jumps, and one halt. The maximum source address length is 36.

The checker uses the inherited register-box interface for arbitrary natural counters. It does not expand every arithmetic-box operation into an unbounded concrete trace. Its proof interpretation is a finite composition of those unchanged box lemmas with checked literal transitions. Independent V8 implementations reproduced the published primary positive-control receipt and agreed on the new machines. This gives strong executable evidence for the new composition; it is not a newly checked proof-assistant theorem for the whole pipeline.

5.3 Literal blank startup

A separate concrete simulation starts with an empty support set, head zero, and native control 74. Both new machines reach the same complete clean frame after exactly 58,004 steps:

$$q = 84, \qquad \text{head} = -2, \qquad \text{source ordinal} = 25, \quad \text{address} = 100000.$$

The nine physical registers are

$$[0, 0, 0, 0, 0, 0, 2, 0, 0],$$

followed by 26 spare blocks, each of value one. There are 35 blocks altogether and 66 one cells in the entire tape. The complete finite support is included in `evidence/startup_support.json`. The clean frame agrees with the primary frame at step 58,154. This is exact equality of the full tape and control, not a comparison of a displayed tape window.

Future spare-tail lengths can differ between machines because their primitive traces differ. The simulation relation tracks each complete append-one spare file separately. It does not identify different spare tails by fiat.

6 Calibration to PA and the no-push protocol

6.1 Inherited numerical calculus

The retained source uses Cantor pairing

$$p(a,b) = \frac{(a+b)(a+b+1)}{2} + b$$

and constructors

$$I(a,b) = 4p(a,b), \quad M(a,b) = 4p(a,b)+1, \quad U(i,a) = 4p(a,i)+2, \quad E(a,b) = 4p(a,b)+3.$$

Code zero is truth and code one is $M(0,0)$, the designated contradiction. The stack constructor is $[a;w] = p(a,w)$; in particular, $p(0,0) = 0$. The exceptional equality $I(0,0) = 0$ also has to be handled by a fixed truth-proof template. Neither encoding is freely injective into a disjoint truth constructor at zero.

The separate diagonal head is

$$\Gamma(B) = I(B, U(B,B)).$$

Every variable label occurring in the formula coded by B is below B, so the binder in this head is fresh. The primary calibration supplies primitive-recursive proof translations between the separate-head and bundled-gate calculi, and between this calculus's refutations and PA refutations. Its finite binary-membership interpretation handles two-way membership rewriting, the equality witness, and the coincident-variable cases explicitly. The full retained calibration fragment is in `reference/pa194_calibration.tex`.

For soundness, $M(a,b)$ is interpreted as binary membership $\mathrm{Bit}(x_a, x_b)$, and $E(a,b)$ as equality. PA proves finite binary comprehension by induction. Substituting a formula into that induction pattern is a syntactic proof-producing operation; no arithmetical truth predicate or reflection principle is used. For completeness, the supplied proof compiler recovers the original PA196 interfaces using diagonal vacuity and pointed separation. These are inherited logical dependencies, not discoveries of this controller change.

6.2 The corrected direct no-push interface

On a logical command store $(T,W,D,H) = (t,w,d,h)$ put $y = d+x$. The no-push MP command satisfies

$$\mathsf{MP}_x(t,w,d,h) = \begin{cases} (y,w,y,0), & I(t,y) = h, \\ (0,w,y,0), & I(t,y) \neq h. \end{cases}$$

In particular, $D = y$ after a successful MP. Treating the successful branch as if D were zero gives an incorrect compiler argument. The original-guard Shift halts if $t+h = 1$; otherwise its interface is

$$\mathsf{S}(t,w,d,h) = (h, \mathrm{tl}(w), d, \mathrm{hd}(w)).$$

NS has exactly this command interface by the preceding symmetry lemma.

Let $A_0 = \mathsf{AX}(0,0,0)$. The unchanged ten-entry AX interface has output

$$(g, [q_1, \ldots, q_9, t; w], 0, 0),$$

where the selected quantifier head is $q_9 = U(d,h) \geq 2$. The gate and the first eight generated heads have residue zero modulo four; the ninth is $U(d,h)$ and has residue two. Thus generated

heads and the gate are not code one. The saved old top t is not included in that assertion. Useful normalized words are

$$\begin{aligned}\mathsf{Push} &= A_0\,\mathsf{S}^9\,\mathsf{MP}_0,\\ \mathsf{Pop} &= A_0\,\mathsf{S}^{11},\\ \mathsf{Discard} &= \mathsf{MP}_0 \quad \text{when } D = 0.\end{aligned}$$

Their effects are respectively

$$(t, w, d, h) \mapsto (0, [t; w], 0, 0), \qquad (t, [a; w], d, h) \mapsto (t, w, 0, a),$$

and, with $D = 0$,

$$(t, w, 0, h) \mapsto (0, w, 0, 0).$$

The last effect holds on either branch since the MP candidate is zero. The Pop word is total even for top $t = 1$: its last sum guard has the form $U(d, h) + t \geq 2$, rather than comparing t against a zero cache.

Generalization at index i, from $(a, w, 0, 0)$, can use

$$\mathsf{Push}\,\mathsf{MP}_i\,\mathsf{S}\,\mathsf{AX}(i, 0, 0)\,\mathsf{S}^{10},$$

producing $(U(i, a), w, 0, 0)$, or an earlier acceptable halt if a contradiction has already been exposed.

For a modus-ponens proof node with minor conclusion a, major conclusion $I(a, b)$, and target b, an explicit reserved-zero compiler word is

$$\mathsf{Discard}; \mathsf{Push}; C_{\text{major}}; \mathsf{Push}; C_{\text{minor}}; \mathsf{Push}; \mathsf{S}; \mathsf{S}; \mathsf{MP}_b; \mathsf{Pop}.$$

Before the two Shifts its store is

$$(0, [a, I(a, b), 0; w], 0, 0).$$

The Shifts give $(a, [0; w], 0, I(a, b))$, unless a contradiction triggers an acceptable earlier halt. MP then gives $(b, [0; w], b, 0)$, and Pop gives $(b, w, 0, 0)$. The reserved zero is supplied by the compiler, not by the modified successful MP command. This is also valid under the canonical zero-stack convention $[0; 0] = 0$.

For completeness, the axiom leaves also have explicit clean-store selectors. The inductive compiler contract allows every initial top t and stack w, but assumes $D = H = 0$:

$$C_p(t, w, 0, 0) \text{ either halts or returns } (f, w, 0, 0)$$

when p derives f. Thus $\mathsf{Discard} = \mathsf{MP}_0$ is used only with $D = 0$. The all-store wrapper A_0 establishes that precondition.

Number the nine generated heads in their actual stack order. For $j = 1, \ldots, 7$ the head selector is

$$\mathsf{Discard}; \mathsf{AX}_{A,D,H}; \mathsf{S}^{j+1}; \mathsf{Pop}^{9-j}.$$

The saved old top is zero after Discard, so the final Pop leaves a clean cache. The restricted implicational head is

$$\rho(a, b, c, u) = I\big(I(I(c, u), a),\ I(I(a, c), I(b, c))\big), \quad u \leq a.$$

It is selected by

$$\mathsf{Discard}; \mathsf{MP}_u; \mathsf{AX}_{b,a-u,c}; \mathsf{S}^9; \mathsf{Pop}.$$

For a guarded gate $g(a, b, c)$, the selector is

$$\mathsf{Discard}; \mathsf{AX}_{c,b,a}; \mathsf{Pop}^{10}.$$

Only legitimate guarded axiom instances enter this proof-leaf case. Together with the displayed generalization and MP words, these selectors give the structural proof induction without assuming that successful MP still pushes a zero.

These axiom selectors and inductive proof constructors yield a clean proof compiler C_p. Prefixing by A_0 normalizes an arbitrary carried store, and a final Shift exposes contradiction code one. Thus a PA refutation yields a finite halting command word from every carried store, not just from a hypothetical reset. The inherited fair command-word scheduler eventually executes a witnessing word.

Conversely, annotate top, cache, and every live canonical stack entry with PA proofs of their translated formula. AX, MP, and Shift preserve these annotations. The no-push MP omits the extra truth push; it creates no theorem lacking a proof. A sum-guard halt has $(t, h) = (1, 0)$ or $(0, 1)$, exposing a proof of the designated contradiction. The annotations and the canonical zero cases give a primitive-recursive translation from a finite source halt to a PA refutation.

6.3 The resulting theorem and its formal boundary

Theorem 2. *With the repository's retained arithmetic-box lemmas and its explicit separate-head PA calibration, the constructions give*

$$\mathsf{PA} \vdash \neg H_{\mathrm{SUMS}} \longleftrightarrow \mathrm{Con}(\mathsf{PA}), \qquad \mathsf{PA} \vdash \neg H_{\mathrm{NS}} \longleftrightarrow \mathrm{Con}(\mathsf{PA}).$$

Proof. For SUMS, compose its finite-trace equivalence with the primary calibrated source. For NS, compose with the no-push source calibration, equivalently use the corrected direct command compiler above. The universal physical contracts, local quotient witnesses, and literal blank entry transport each source halting statement to its exact binary TM. Finally negate the existential PA-refutation predicate. □

The qualifier identifies mathematical lemmas being reused; it is not a claim of an additional consistency assumption. The deliverable is an ordinary proof with executable finite certificates. It does not claim that all referenced calibration text, source compilation, box contracts, and literal tables have now been assembled into a new single Lean or Rocq theorem. The repository's stronger iPRA-endpoint questions also remain separate.

7 Attempts below 194 and lessons from rejected reductions

No lower bound is proved. Several finite search spaces and proposed logical shortcuts were examined without producing a certified smaller table.

A direct scanner merge $34 \to 75$ fails 113 primary symbolic outcomes, at 57 source sites. The source has relevant leading-zero paths and transient zero stretches outside a naive AX-only bound. Existing six-state synthetic scanner examples do not validate that merge for the real source.

The surveyed 229 retained valid grammar objects did not give a usable maximum zero stretch of five. Expanded history-context searches, recursive equal-branch pair-fold searches, and scanner merges into non-engine controls likewise produced no accepted 193-state table. These are failures of the specified searches, not a proof that 193 states are impossible. Some historical packets already contain very large grammar searches; rediscovering their negative candidates is not new progress.

A proposed swap of the classical head $\neg B \to H$ into $\neg H \to B$ was rejected: a derivation using double-negation elimination or explosion is circular if those principles themselves rely on

the old head. Curried-premise exchange in safe implicational/quantifier heads can be derived using independent propositional templates, but it was already present in the comparison corpus and did not yield a new sub-194 allocation here.

A tempting deletion of the second membership-rewrite head is also invalid. Interpret $M(s, z)$ as the z-th binary bit of s, and $E(s, t)$ as inclusion of the corresponding finite sets. Container transport and the equality-witness head remain valid. But the proposed deleted head

$$E(c, a) \to (M(b, c) \to M(b, a))$$

fails for distinct variable labels a, b, c under the valuation $x_c = 0$, $x_a = x_b = 1$: $E(c, a)$ and $M(b, c)$ are true, while $M(b, a)$ is false. The designated contradiction $M(0, 0)$ is still false. Thus this does not merely exploit a changed meaning of falsehood. It refutes the proposed derivability shortcut; it does not preclude every possible different PA interpretation of a reduced calculus.

The actionable outcome is the sum-comparator transfer elimination and its compensating history fold. Further state reduction must pay for an actual source protocol and pass complete physical contracts; a small source expression or a plausible local scan bound does not suffice.

8 Artifacts, reproduction, and limitations

The native and dense tables are complete, with one five-column row `source read write L-or-R target` per read/control pair. The following hashes fix their exact bytes (LF endings and a final LF).

NS native
`db2ce84a0e883b44e348b60c834382245a9f55623de675ffdacb1e8b3c18bf75`

NS dense
`eef9329b9a7e1f2810eddcf3e99fc0184b073ea266d658ebe473f7d11d68ecd7`

SUMS native
`b81a833dc99bcc2204ba513260e2e4fa07981b018228eef2f0bcc188e977f9c2`

SUMS dense
`f7c8799c86847a4aeb85f078015f159a6fabbc94d5e26cbd84f337c4f8789fa8`

The source bundle retains the primary compiler and symbolic runtime unmodified, the pinned upstream source DAG, grammar and folds, both new tables and label maps, the checked receipts, complete startup support, and the James Chang MIT license. All original contributions are credited through the pinned source paths.

With Python 3.10 or later, run

```
python3 verify.py
```

The script rebuilds the raw parents and quotients, checks all eight folds, checks state count and hashes, runs the primary positive control and both new symbolic suites, and checks complete concrete startup. Assertions are part of the verification; do not use `python -O`. The alternatives

```
node verify.mjs
node audit/verify_audit.mjs
```

run the separate JavaScript source checker and the complete 51-table graph audit under Node.js 18 or later. The core algorithm was executed in V8, and both Node.js file-I/O wrappers were executed successfully after the workspace became available. To build this report:

```
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The supplied reconstruction script writes the preserved files into a new directory, verifies the Python replay, requires a working pdflatex, compiles twice, and then creates the ZIP. The complete Python replay, independent Node.js source checks, and Node.js graph audit passed. The report was compiled and its pages visually inspected before delivery.

The graph audit establishes novelty relative to the retrieved pinned corpus; it is not a worldwide priority search. The unsuccessful reduction searches establish no state lower bound. The numerical calibration is an inherited mathematical proof, and the arithmetic-box interface is an inherited lemma dependency. The checked finite certificates do not replace those logical facts.

Primary sources

All repository paths below are relative to https://github.com/VladimirReshetnikov/ProveIt/tree/1e407ae841e4666f932777fad977ba784ebbcfd5.

1. `Computability/BusyBeaver/Research/PA/README.md` and the `docs/README.md` inventory.
2. `Computability/BusyBeaver/Research/PA/docs/pa194_conpa_2026-11-13/`: primary tables, grammar, folds, symbolic runtime, and calibration.
3. `Computability/BusyBeaver/Research/PA/jamesyc-upstream/pa196/`: source DAG, M6 arithmetic interface, and retained license.
4. `docs/incoming/pa194N_conpa_2026-11-13.zip`: no-push source and its mathematical calibration.
5. Incoming families and diagnostics:
   `docs/incoming/PA194R_research_2026-11-14.zip`
   `docs/incoming/pa194_radix_family_2026-11-13.zip`
   `docs/incoming/pa194j_pa194d_conpa_2026-11-13.zip`
   `docs/incoming/pa194q_conpa_2026-11-13.zip`.
6. `Computability/BusyBeaver/Research/PA/docs/pa194_new_designs_2026-11-13/`: cached-head symmetric comparison.
7. `Computability/BusyBeaver/Research/PA/docs/pa195_followup_research/`: scanner argument and failed shortcuts.
8. Related research:
   `Computability/BusyBeaver/Research/ZFC/`
   `Logic/PrimitiveRecursiveArithmetic/ConciseIPRA/`.

A Packet ledger

No.	Packet	Assessment
1	`pa_unary_power_cover`	Baseline unary-power construction; contextual folding reference
2	`U208_ConPA_Conditional_Context_2026-10-29`	208; conditional context
3	`U208_ConPA_Entry_Context_2026-10-29`	208; entry context
4	`U208_ConPA_History_2026-10-29`	208; history folding
5	`U208_ConPA_2026-10-29`	208; three-cell contexts
6	`u208_research_2026-10-29`	Two 208 constructions

No.	Packet	Assessment
7	Below_U208_Research_2026-10-29	No certified table at 207 or below
8	u208_comparison_research_2026-10-29	Comparison; no below-208 construction
9	ConPA_below208_research_2026-10-29	No 207-state result
10	pa_consistency_research_2026-10-29	No below-208 result
11	pa208_research_2026-10-29	No below-208 result; historical 27 presentations / 20 graphs
12	pa_tm_research_2026-11-13	Shorter source paths; 196
13	pa_consistency_research_2026-11-13	Diagonal vacuity; 199
14	conpa_tm_comparison_and_reduction_search	Two 196 command variants; prior no-push ideas
15	pa196_research_2026-11-13	157-node source; no below-196 table
16	pa_consistency_research	No below-196 construction
17	pa_tm_comparison_and_reductions	Source rewrite research
18	pa_tm_extended_research_2026-11-13	No below-196 construction
19	conpa_tm_comparison_and_search	AD layout 196; scanner failures; rejected 195
20	pa_consistency_reduction	Two 196 tables; shorter source compiles to 203
21	pa_consistency_comparison_2026_11_13	No below-196 construction
22	PA_consistency_TM_research_2026-11-13	Maximum-zero-five tree unallocated; no below-196 table
23	pa_consistency_research_2026-11-13_2	196/198 layouts; joint register/fold search
24	conpa_tm_study_2026-11-13	No below-196 construction
25	pa_below196_research	No below-196 construction
26	pa_consistency_comparison_and_new196_2026-11-13	New 196; no below-196 table
27	pa_tm_research_2026-11-13_2	Tag and premise-exchange search already present; no below-196 table
28	pa_consistency_cross_archive	No certified 195-state construction
29	pa195_conpa_2026-11-13	Original 195
30	pa_consistency_reduction_audit_2026-11-14	Audit; no new below-196 theorem
31	pa_consistency_research_2026-11-14	NEG18 shorter source but 197-state TM
32	PA195_ConPA_research_2026-11-14	195S separate; 195C cached
33	pa195_followup_research	M195S8; longer scanner; no below-195 result
34	pa194_conpa_2026-11-13	Primary 67/7 construction and AD companion
35	pa194_conpa_2026-11-13_2	Alternative 68/8 construction
36	pa195_diagonal_combined_2026-11-13	195DC combined diagonal
37	pa194C_conpa_2026-11-13_2	Cached C duplicate
38	pa194_new_designs_2026-11-13	C and cached symmetric CS
39	pa194c_conpa	Cached C; supplementary cached 68/8
40	pa194c_conpa_2026-11-13_1	Cached C and CZ zero jumps
41	pa194c_conpa_2026-11-13	Cached C; large scanner search
42	PA194R_research_2026-11-14.zip	R, F, A and X; incoming
43	pa194N_conpa_2026-11-13.zip	No-push N with original sum guard; incoming
44	pa194_radix_family_2026-11-13.zip	Radix 4R, 5, 6; incoming
45	pa194j_pa194d_conpa_2026-11-13.zip	J, D and grammar siblings; incoming
46	pa194q_conpa_2026-11-13.zip	Q premise exchange; CSZ duplicate; incoming

B Prior 194-state graph classes

The digest prefixes identify representative dense table bytes. They are not the canonical graph digests. Exploration and rejected candidates are included for novelty comparison, not promoted to consistency theorems.

No.	Representative	Dense SHA prefix	Classification
1	Primary 67/7	`4bfea62b17aa`	Established primary construction
2	AD register layout	`27613aa92da7`	Register-allocation companion
3	Alternative 68/8	`9087d70fc0e1`	Alternative AX quotient
4	Scanner candidate	`c48d6711e9c7`	Rejected diagnostic
5	Cached C	`d759313c03b5`	Five exact duplicate deliveries
6	Cached symmetric CS	`1c71002e5b8a`	Cached-head symmetry
7	Cached 68/8	`a509952c1e06`	Supplementary allocation
8	CZ	`b713e73457be`	Zero-jump variant
9	N	`37f7e909724c`	No-push original-guard protocol
10	A	`3c3e74ffcb5d`	Incoming A variant
11	F	`4fcee0c2ef7b`	Incoming F variant
12	R	`8a9a8a30773f`	Incoming R variant
13	X = CSZ	`62e93098caf7`	Exact shared graph across incoming R/Q/radix
14	103	`55ef6a817807`	Rejected diagnostic
15	bothzero6	`28b3a48e3485`	Exploratory table
16	MPH	`f668a2f4de78`	Search table
17	Radix 4R	`a3c1d556d61b`	Coefficient-4 decoder layout
18	Radix 5	`180a0fd1f046`	Formula coefficient 5
19	Radix 6	`725ce8474d7b`	Formula coefficient 6
20	C_I_Z	`1dd4bf6a2c6b`	Search table
21	node1X	`f8abcd947786`	Rejected diagnostic
22	J	`7c498ced0eb0`	Zero-jump construction
23	D	`4226b7702de5`	Diagonal/gate construction
24	gammaDCSZA	`f9527c1867a0`	Incoming J/D sibling
25	finalA3	`57a1edbaafb0`	Incoming J/D sibling
26	finalA5	`2bcebbeefd95`	Incoming J/D sibling
27	finalA6	`4bccc41f2e84`	Incoming J/D sibling
28	Q	`48e98b5504dc`	Premise exchange; invariant presentation same graph

C Complete dense PA194NS table

State	Read 0: write, move, next	Read 1: write, move, next
0	$(1, R, 1)$	$(1, R, 2)$
1	$(1, R, 3)$	$(1, R, 1)$
2	$(0, R, 0)$	$(1, R, 2)$
3	$(0, R, 4)$	$(0, R, 1)$
4	$(1, R, 5)$	$(1, L, 6)$
5	$(1, L, 7)$	$(1, L, 8)$
6	$(0, L, 7)$	$(1, L, 6)$
7	$(0, L, 9)$	$(1, L, 6)$
8	$(0, L, 10)$	$(0, L, 11)$
9	$(1, R, 12)$	$(0, L, 9)$
10	$(0, L, 9)$	$(0, L, 10)$
11	$(1, L, 8)$	$(1, L, 11)$
12	$(0, R, 12)$	$(0, R, 13)$
13	$(0, L, 14)$	$(0, R, 13)$

State	Read 0	Read 1
14	$(1, L, 15)$	$(0, L, 15)$
15	$(0, L, 15)$	$(1, L, 16)$
16	$(0, L, 17)$	$(1, L, 16)$
17	$(0, L, 18)$	$(1, L, 16)$
18	$(0, L, 19)$	$(1, L, 16)$
19	$(0, L, 20)$	$(1, L, 16)$
20	$(0, L, 21)$	$(1, L, 16)$
21	$(0, L, 22)$	$(1, L, 16)$
22	$(0, L, 23)$	$(1, L, 16)$
23	$(0, R, 23)$	$(1, R, 24)$
24	$(0, R, 25)$	$(1, R, 26)$
25	$(0, R, 27)$	$(1, R, 28)$
26	$(0, R, 29)$	$(1, R, 30)$
27	$(0, R, 31)$	$(1, R, 194)$
28	$(0, L, 32)$	$(1, R, 33)$
29	$(0, R, 34)$	$(1, R, 35)$
30	$(0, L, 33)$	$(1, L, 36)$
31	$(0, R, 37)$	$(1, R, 38)$
32	$(0, R, 39)$	$(1, R, 40)$
33	$(0, R, 41)$	$(1, R, 42)$
34	$(0, R, 43)$	$(1, R, 44)$
35	$(0, R, 45)$	$(1, R, 46)$
36	$(0, R, 47)$	$(1, R, 48)$
37	$(0, R, 49)$	$(1, R, 41)$
38	$(0, R, 50)$	$(1, R, 51)$
39	$(0, R, 52)$	$(1, R, 53)$
40	$(0, R, 54)$	$(1, R, 55)$
41	$(0, R, 56)$	$(0, R, 57)$
42	$(0, R, 58)$	$(0, L, 14)$
43	$(0, R, 59)$	$(1, R, 60)$
44	$(0, R, 52)$	$(1, R, 61)$
45	$(0, L, 62)$	$(1, R, 63)$
46	$(0, R, 64)$	$(1, R, 65)$
47	$(0, R, 66)$	$(0, R, 67)$
48	$(0, L, 64)$	$(1, L, 68)$
49	$(0, R, 69)$	$(1, R, 51)$
50	$(0, L, 70)$	$(0, L, 15)$
51	$(0, R, 71)$	$(0, L, 10)$
52	$(0, R, 72)$	$(1, R, 73)$
53	$(0, R, 74)$	$(1, R, 75)$
54	$(0, R, 76)$	$(1, R, 77)$
55	$(0, R, 78)$	$(1, R, 79)$
56	$(0, L, 80)$	$(0, R, 1)$
57	$(0, R, 81)$	$(1, R, 57)$
58	$(0, L, 72)$	$(1, L, 65)$
59	$(0, R, 82)$	$(0, L, 7)$
60	$(0, R, 83)$	$(1, R, 84)$
61	$(0, R, 41)$	$(1, R, 85)$
62	$(0, R, 86)$	$(1, R, 87)$
63	$(0, R, 88)$	$(0, L, 71)$
64	$(0, R, 30)$	$(1, R, 87)$
65	$(0, R, 69)$	$(1, R, 36)$
66	$(0, L, 80)$	$(0, R, 2)$
67	$(0, R, 89)$	$(1, R, 67)$
68	$(0, R, 90)$	$(1, R, 91)$
69	$(0, R, 56)$	$(0, R, 81)$
70	$(0, R, 65)$	$(1, R, 92)$

State	Read 0	Read 1
71	$(0, R, 41)$	$(0, L, 13)$
72	$(0, R, 82)$	$(1, R, 68)$
73	$(0, R, 48)$	$(1, R, 93)$
74	$(0, R, 41)$	$(1, R, 94)$
75	$(0, R, 95)$	$(1, R, 96)$
76	$(0, R, 97)$	$(1, L, 98)$
77	$(0, R, 48)$	$(1, R, 76)$
78	$(0, R, 92)$	$(1, R, 99)$
79	$(0, R, 48)$	$(1, R, 100)$
80	$(1, R, 80)$	$(1, L, 15)$
81	$(0, R, 101)$	$(1, R, 81)$
82	$(0, L, 45)$	$(0, R, 89)$
83	$(0, R, 102)$	$(1, R, 93)$
84	$(0, R, 102)$	$(1, R, 103)$
85	$(0, R, 104)$	$(0, L, 14)$
86	$(0, R, 105)$	$(1, L, 70)$
87	$(0, R, 103)$	$(0, L, 14)$
88	$(0, R, 106)$	$(1, R, 107)$
89	$(0, R, 108)$	$(1, R, 89)$
90	$(0, R, 66)$	$(0, R, 109)$
91	$(0, R, 110)$	$(0, L, 87)$
92	$(0, R, 30)$	$(1, R, 58)$
93	$(0, R, 82)$	$(1, L, 111)$
94	$(0, R, 99)$	$(0, L, 14)$
95	$(0, R, 64)$	$(1, R, 112)$
96	$(0, R, 111)$	$(1, R, 74)$
97	$(0, L, 45)$	$(0, R, 108)$
98	$(0, L, 44)$	$(1, R, 113)$
99	$(0, L, 114)$	$(1, R, 115)$
100	$(0, R, 116)$	$(0, L, 14)$
101	$(0, R, 109)$	$(1, R, 101)$
102	$(0, L, 45)$	$(0, R, 101)$
103	$(0, L, 114)$	$(1, L, 33)$
104	$(0, R, 66)$	$(0, R, 117)$
105	$(0, R, 56)$	$(0, R, 117)$
106	$(0, L, 62)$	$(1, R, 118)$
107	$(0, R, 119)$	$(1, R, 120)$
108	$(0, R, 14)$	$(1, R, 108)$
109	$(0, R, 117)$	$(1, R, 109)$
110	$(0, R, 71)$	$(1, R, 121)$
111	$(0, R, 122)$	$(1, L, 123)$
112	$(0, R, 124)$	$(1, R, 125)$
113	$(0, R, 47)$	$(0, L, 14)$
114	$(0, R, 90)$	$(1, R, 98)$
115	$(0, R, 114)$	$(1, L, 93)$
116	$(0, R, 126)$	$(1, R, 126)$
117	$(0, R, 127)$	$(1, R, 117)$
118	$(0, R, 128)$	$(1, R, 129)$
119	$(0, R, 32)$	$(1, R, 64)$
120	$(0, R, 31)$	$(1, R, 96)$
121	$(0, R, 48)$	$(1, R, 87)$
122	$(0, R, 56)$	$(0, R, 127)$
123	$(0, R, 65)$	$(1, R, 33)$
124	$(0, R, 97)$	$(1, R, 113)$
125	$(0, R, 78)$	$(1, R, 77)$
126	$(0, R, 47)$	$(1, R, 47)$
127	$(0, R, 67)$	$(1, R, 127)$

State	Read 0	Read 1
128	$(0, R, 130)$	$(1, R, 131)$
129	$(0, R, 132)$	$(1, R, 133)$
130	$(0, R, 134)$	$(1, R, 135)$
131	$(0, R, 136)$	$(1, R, 137)$
132	$(0, R, 138)$	$(1, R, 139)$
133	$(0, R, 140)$	$(1, R, 141)$
134	$(0, R, 95)$	$(1, R, 123)$
135	$(0, R, 142)$	$(1, R, 126)$
136	$(0, R, 143)$	$(1, R, 144)$
137	$(0, R, 145)$	$(1, R, 146)$
138	$(0, R, 134)$	$(1, R, 147)$
139	$(0, R, 47)$	$(1, R, 148)$
140	$(0, R, 149)$	$(1, L, 131)$
141	$(0, R, 150)$	$(1, R, 151)$
142	$(0, R, 75)$	$(1, R, 40)$
143	$(0, R, 28)$	$(1, R, 152)$
144	$(0, L, 153)$	$(1, L, 154)$
145	$(0, L, 129)$	$(1, R, 155)$
146	$(0, R, 156)$	$(1, R, 157)$
147	$(0, R, 149)$	$(1, R, 158)$
148	$(0, R, 159)$	$(1, R, 47)$
149	$(0, R, 95)$	$(1, R, 160)$
150	$(0, R, 161)$	$(1, R, 162)$
151	$(0, R, 163)$	$(0, L, 10)$
152	$(0, R, 164)$	$(1, R, 165)$
153	$(0, R, 166)$	$(1, R, 157)$
154	$(0, R, 167)$	$(1, R, 156)$
155	$(0, R, 135)$	$(1, R, 130)$
156	$(0, R, 147)$	$(1, R, 40)$
157	$(0, R, 134)$	$(1, R, 168)$
158	$(0, R, 40)$	$(1, L, 169)$
159	$(0, L, 170)$	$(1, R, 171)$
160	$(0, R, 86)$	$(1, R, 61)$
161	$(0, R, 172)$	$(1, R, 170)$
162	$(0, R, 173)$	$(1, R, 174)$
163	$(0, R, 175)$	$(1, R, 176)$
164	$(0, R, 177)$	$(1, R, 178)$
165	$(0, L, 179)$	$(1, L, 180)$
166	$(0, R, 149)$	$(1, R, 140)$
167	$(0, R, 165)$	$(1, R, 181)$
168	$(0, L, 130)$	$(1, R, 47)$
169	$(0, R, 182)$	$(1, R, 135)$
170	$(0, R, 183)$	$(1, R, 145)$
171	$(0, R, 158)$	$(1, R, 130)$
172	$(0, R, 139)$	$(1, R, 165)$
173	$(0, R, 158)$	$(1, R, 184)$
174	$(0, R, 185)$	$(1, R, 86)$
175	$(0, R, 31)$	$(1, R, 50)$
176	$(0, R, 186)$	$(1, R, 175)$
177	$(0, R, 134)$	$(1, R, 142)$
178	$(0, R, 32)$	$(1, R, 187)$
179	$(0, R, 144)$	$(1, R, 188)$
180	$(0, R, 189)$	$(1, R, 190)$
181	$(0, R, 40)$	$(1, R, 191)$
182	$(0, R, 168)$	$(1, R, 40)$
183	$(0, R, 192)$	$(1, R, 47)$
184	$(0, R, 153)$	$(1, R, 146)$

State	Read 0	Read 1
185	$(0, R, 165)$	$(1, R, 180)$
186	$(0, R, 121)$	$(1, R, 111)$
187	$(0, R, 193)$	$(1, R, 159)$
188	$(0, R, 169)$	$(1, R, 154)$
189	$(0, R, 158)$	$(1, R, 192)$
190	$(0, R, 179)$	$(1, R, 182)$
191	$(0, R, 182)$	$(1, R, 40)$
192	$(0, R, 149)$	$(1, R, 168)$
193	$(0, R, 75)$	$(1, R, 144)$